Deep-Dive Analysis & Interactive Dashboard

ApexPlanet Software Pvt. Ltd.

Data Analytics Internship

Task 3

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Objective:

The objective of this project is to analyse sales performance using Power BI and develop an interactive dashboard that helps users understand sales trends, customer behaviour, product performance and regional revenue. The dashboard enables business users to filter data dynamically and make informed decisions based on real-time insights.

Dataset Overview:

The sales dataset contains transactional records including customer information, product details, sales amount, quantity purchased, city, gender, and order date. The dataset was cleaned and prepared before being imported into Power BI for dashboard development.

Data Cleaning:

Before performing analysis, the dataset was inspected for quality issues.

Data Quality Checks:

- Missing values were identified.
- Duplicate records were checked.
- Data types were verified.
- Date format was standardized.

Dashboard Overview:

The Sales Analysis Dashboard was developed to provide business users with an interactive view of sales performance. The dashboard combines KPI cards, charts, customer segmentation, and slicers to help users analyse revenue trends, product performance, customer behaviour, and geographical sales distribution.

Dashboard Features:

- KPI Cards
- Revenue by City
- Revenue by Product Category
- Monthly Sales Trend
- Top 10 Products
- Customer Segmentation
- Gender Distribution
- Interactive Filters (Category, City, Month, Gender)

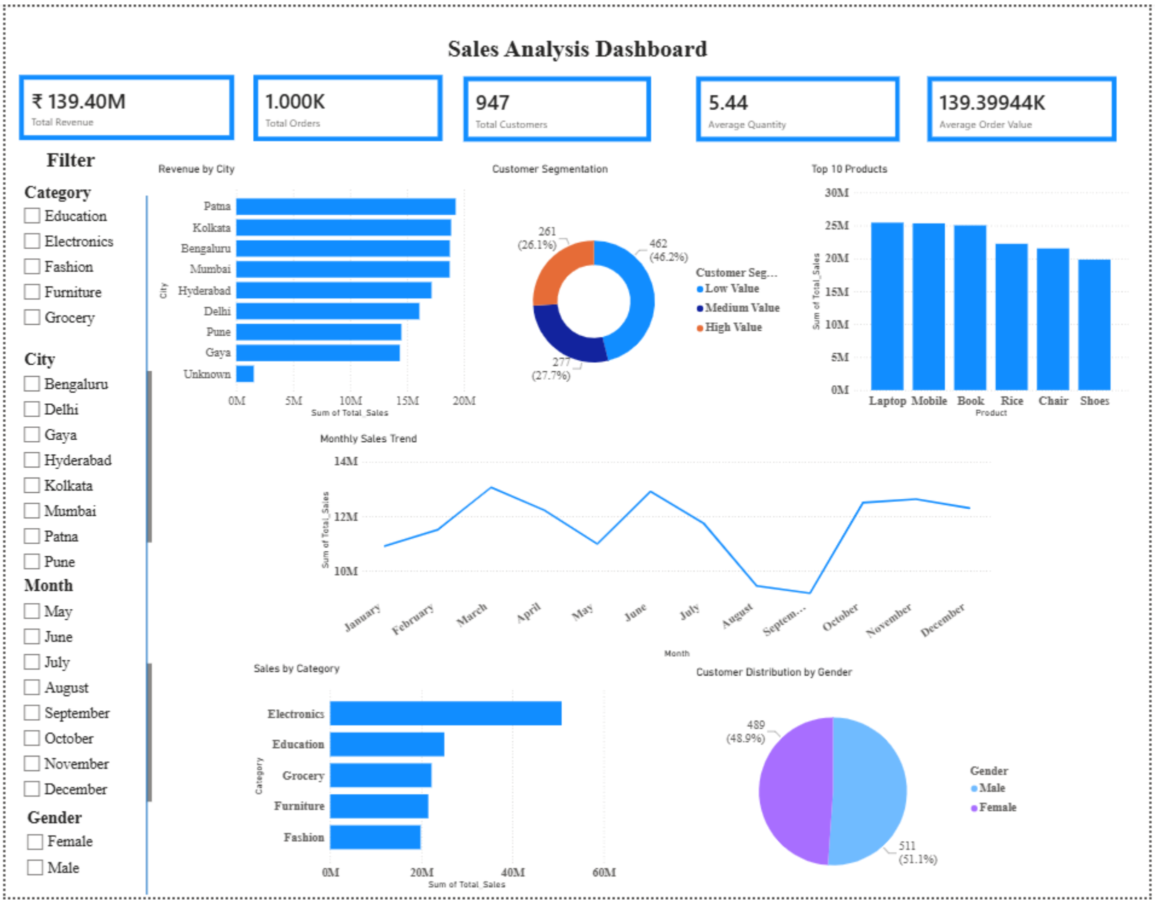
Deep-Dive Analysis:

Customer segmentation was selected as the focus area of the deep-dive analysis. Customers were grouped into High Value, Medium Value, and Low Value segments based on their total sales value.

This segmentation enables businesses to:

- Identify high-value customers for loyalty programmes.
- Target medium-value customers with promotional campaigns.
- Develop strategies to improve engagement among low-value customers.

The dashboard allows users to analyse these customer groups interactively using filters.



KPIs:

Total Revenue

Represents the overall revenue generated from all customer orders.

Total Orders

Displays the total number of orders placed.

Total Customers

Shows the number of unique customers.

Average Order Value

Average revenue generated per customer order.

Average Quantity

Average quantity of products purchased in each order.

Dashboard Insights:

Key Findings

- Electronics generated the highest revenue.
- Laptop is the highest-selling product.
- Patna generated the highest sales among all cities.
- Male and female customer distribution is almost equal.

- Most customers belong to the Low Value customer segment.
- Monthly sales show fluctuations, indicating seasonal demand.

Business Recommendations:

- Increase inventory for Electronics products.
- Focus marketing campaigns on top-performing cities.
- Provide loyalty programmes for high-value customers.
- Promote low-performing product categories.
- Monitor monthly sales trends for inventory planning.

Conclusion:

This project successfully demonstrates the use of Microsoft Power BI for analysing sales performance through an interactive dashboard. The dashboard provides meaningful insights into customer behaviour, product performance, regional sales, and monthly trends. By integrating KPI monitoring, customer segmentation, and interactive filtering, the solution supports data-driven business decision-making and highlights the value of business intelligence in modern organisations.